



De-Havilland DHC-8-400 B1 & B2 TRAINING MANUAL

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CHAPTER ATA 79 ENGINE OIL

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Oil Distribution System

Introduction

The oil system supplies clean, filtered oil to the engine for lubrication of moving parts and removes any unwanted material.

General Description

The oil system is a wet-sump system with the oil cooled by an aircraft mounted air-cooled oil cooler. The oil system supplies a constant flow of clean filtered oil to the turbomachine and reduction gearbox bearings to cool and lubricate the running shafts, gears and component surfaces. This flow of oil also carries away any contamination. Air that is trapped in the oil is removed from the system by a cyclonic de-aerator, located at the top of the oil pump pack. Filters in the pressure and scavenge systems remove contaminants from the oil. The filter housings contain impending bypass switches, which sense pressure differential. When a filter is becoming blocked, the switches send a signal to the related Full Authority Digital Electronic Control (FADEC) and the Engine Monitoring System (EMS) to warn of an impending bypass.

Detailed Description

The oil used in the system is contained in a tank that is integral to the LP compressor case. The tank has a sight glass, for viewing the oil quantity, and a filler neck and cap for replenishing the oil supply. A second, smaller, tank is located in the reduction gearbox. This tank is supplied with oil from the main tank. The oil pumps are assembled together as one unit and the entire assembly is installed in a bore located on the LP compressor case.

The Oil system consists of the three sub-systems that follow:

- Pressure system that supplies oil to the reduction gearbox and the turbomachinery
- Scavenge system that returns the used oil to the tank
- Vent and breather system that vents the bearing cavities and removes any air trapped in the scavenged oil.

Pressure System

[Refer Figure 79-1](#)

A vane type pressure pump is installed in a pack with the scavenge pumps, on the left side of the Low Pressure (LP) compressor case. Integral passages connect the oil tank to the inlet side of the pump. Oil, under pressure, flows from the pump to the Pressure Regulating Valve (PRV) and to the Air Cooled Oil Cooler (ACOC). A pressure relief valve (cold start valve) returns oil to the tank to prevent a pressure surge during cold engine starting.

The PRV consists of a piston valve and spring in a ported sleeve. The PRV keeps a constant oil pressure above and in relation to the air pressure from the bearing cavities. If oil pump output pressure overcomes the air pressure plus spring pressure, the valve opens a port. Oil is bled from the main pressure line through the port and returned to the inlet side of the pump, reducing the output pressure. Air pressure plus spring pressure overcomes the reduced oil pressure, closing the bleed port at the correct pump output pressure.

Airframe-supplied tubes connect the outlet to the airframe ACOC. From the oil-cooler the oil flows to the pressure oil filter.

On aircraft with ModSum 4-113604 incorporated, the concentric spool is added to the Pressure Regulating Valve (PRV). The PRV maintains the constant oil pressure even in the below zero gravity (g) condition. This is achieved through the concentric spool in the PRV interrupts the oil bypass flow at the above idle condition which eliminates the propeller oil pressure loss. The spring loaded piston at the end of the PRV casing actuates the spool.

The pressure filter removes contaminants from the oil and has a bypass valve to ensure adequate flow if the filter is blocked. From the filter the oil flows to the fuel heater, where it warms the fuel to prevent the formation of ice crystals. From the fuel heater the oil goes to the bearings, air intake anti-icing passages and to the Reduction Gearbox (RGB). Strainers located before the bearings collect any contaminants that may be present in the oil. Oil flows to the RGB auxiliary tank from the turbomachinery. The auxiliary tank is part of

the RGB casting and is pressurized when the engine is running. This tank is always full of oil, even when the engine is not running.

Oil, from the auxiliary tank, is distributed through internal galleries to the reduction and accessory geartrain, bearings and to the ac generator, where it is used for cooling. Oil also flows, by internal passages and tubes, to the components that follow:

- Auxiliary feathering pump
- Overspeed governor
- Pitch Control Unit (PCU) pump

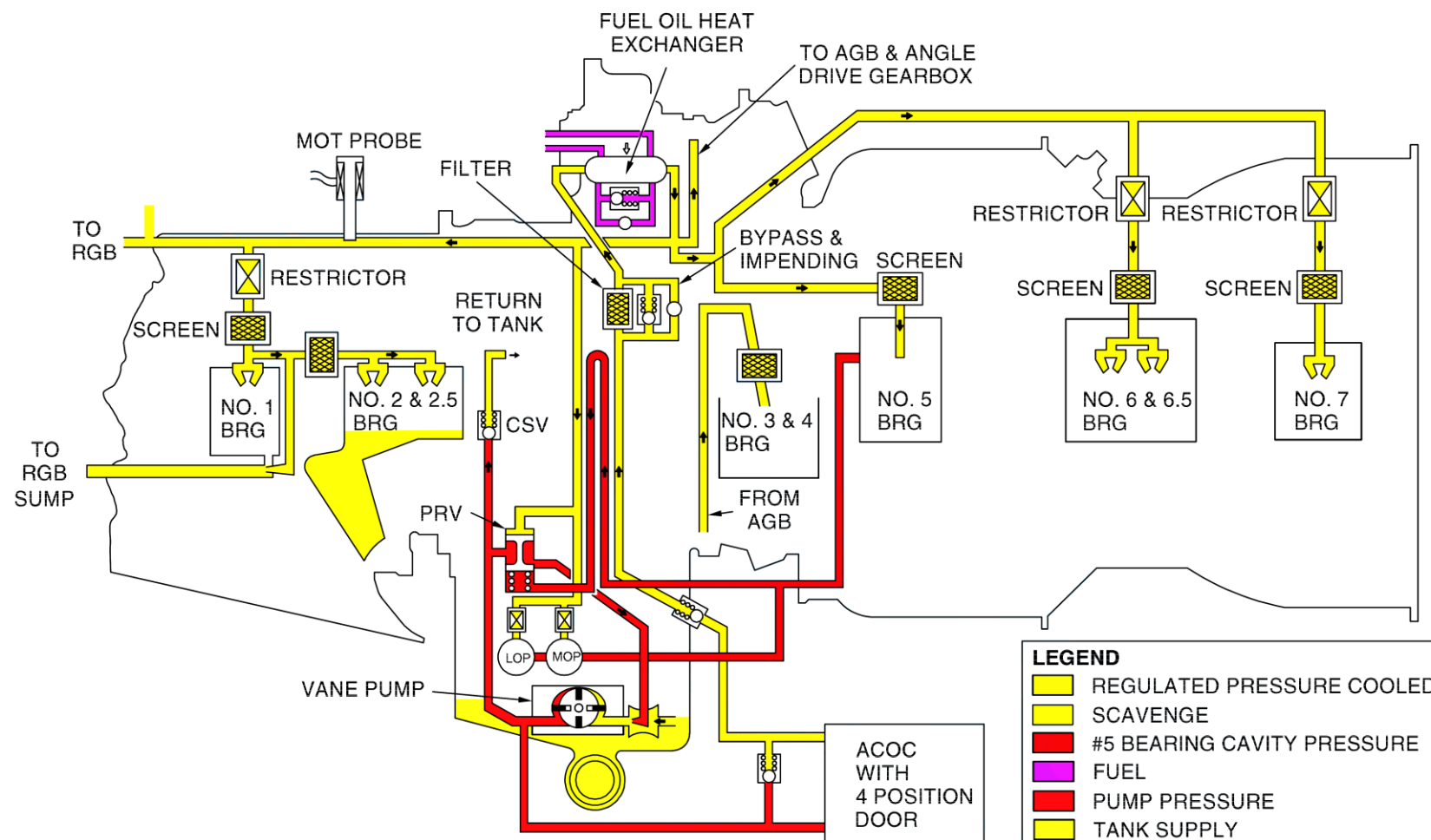


Figure 79-1 Oil Pressure System

Scavenge System

[Refer Figure 79-2](#)

Oil from the No. 1 bearing is scavenged by gravity to the Reduction Gearbox (RGB) cavity. Bearings 2, 2.5, 3, 4, and 5 are scavenged by dedicated scavenge

pumps through internal oil passages, while, bearings 6, 6.5 and 7 are scavenged by dedicated scavenge pumps through external oil tubes. Oil located in the Accessory Gearbox (AGB) geartrain and bearings is also scavenged by a dedicated pump. The scavenged oil is collected and sent to the cyclonic de-aerator located at the top of the pump pack. The oil then returns to the tank.

Scavenge oil from the RGB accessories, gears and bearings drains into a cavity in the bottom of the RGB rear housing. A chip detector is located in this cavity, and traps magnetic contaminants that can be used as an indication of RGB wear. From the RGB cavity, oil flows through the inlet duct lip for anti-icing of the intake and then to the RGB scavenge pump. Oil from the ac generator flows past another chip detector, through a screen and then to the ac scavenge pump, which is part of the pack on the turbomachinery. Oil from the ac and RGB scavenge pumps flows up through the scavenge filter to the cyclonic de-aerator and then back to the tank. The scavenge filter housing is equipped with a valve to bypass oil around the filter in the event of blockage. An indicator warns of an impending bypass.

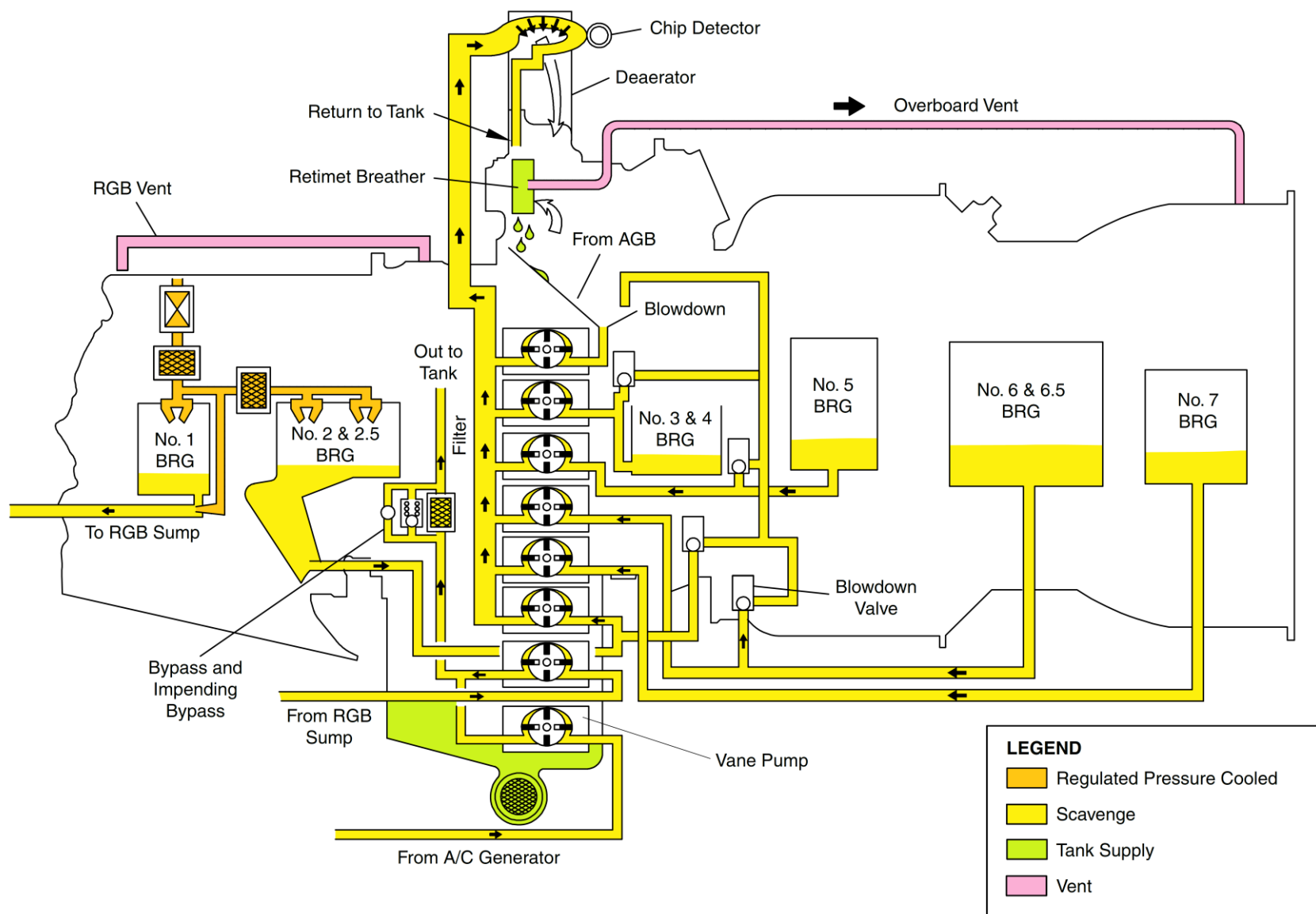


Figure 79-2 Oil Scavenge System

Vent and Breather System

Refer Figure 79-2

A cyclonic de-aerator, located at the top of the pump pack, removes any air that is mixed in with the oil. Oil from the de-aerator drains into the AGB and then back to the oil tank. The cyclonic de-aerator housing contains a chip detector which traps magnetic contaminants. These contaminants can be used as an indication of engine wear.

Refer Figure 79-3

RGB Oil Pressure System

RGB auxiliary oil tank:

- Is located in the RGB rear housing.
- It is a pressurized tank with capacity of 5.3 US quarts (5 litres).

It supplies oil to:

- RGB gear train (for lubrication)
- AC generator (for lubrication)
- PCU pump (to operate the propeller system)
- Auxiliary feathering pump (to feather the propeller in an emergency during take-off).
- The Auxiliary Feathering Pump is the only LRU that can take oil in the tank in static condition.

RGB Oil Scavenge System

RGB scavenge oil filter housing:

- Has a 12 micron filter element.
- Has an impending bypass indicator: at 17–24 psid (117– 165 kPad).
- Has a filter bypass valve that maintains oil supply when filter becomes restricted.
- It opens at 34 psid (234 kPad).

A/C generator:

- Is scavenged by a vane type pump and returns oil to the tank through

the A/C chip detector and RGB scavenge oil filter.

- In case of an overflow, oil is directed to RGB sump.

Propeller system, Overspeed Governor and Gear Train:

- Oil is directed to RGB sump.

RGB sump:

- Is scavenged by a vane type pump and returns oil to the tank across the RGB chip detector and through the scavenge oil filter.
- RGB scavenge oil temperature is also used to warm up the front inlet case rear lip to prevent an ice build-up (antiicing).

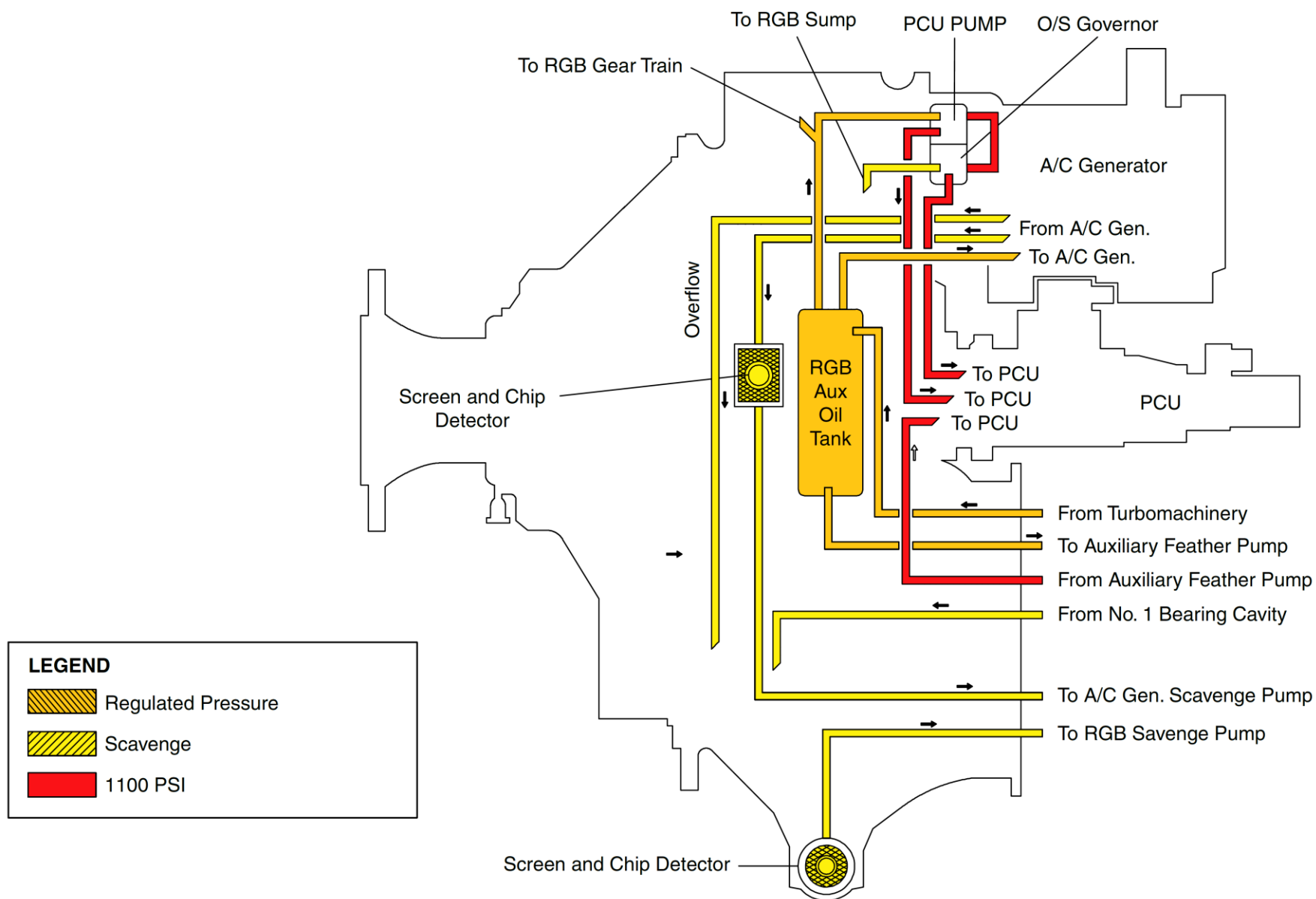


Figure 79-3 RGB Oil System

Oil Cooler System

Introduction

The oil cooler system keeps the engine oil temperature within limits.

General Description

The oil cooler is installed in the oil pressure circuit of the PW150A engines. The assembly is made from the oil cooler unit, the bypass valve, the air inlet duct, and the exit duct. The oil cooler system has the components that follow:

- Oil Cooler
- Oil Cooler Bypass Valve
- Exit Duct
- Oil Cooler Ejector
- Oil Cooler Ejector Valve
- Oil Cooler Air Outlet Flap
- Oil Cooler Air Outlet Flap Actuator

Detailed Description

The oil cooler uses the air from the engine air intake to decrease the heat in the oil, which then releases to the atmosphere.

The air inlet duct is welded to the oil cooler and is shaped to let the airflow go through it. The exit duct is not part of the assembly, but is supplied with bolts and anchor nuts to attach it. The oil cooler is attached with a twelve bolt/anchor nut configuration between two brackets at the engine firewall structure.

Oil Cooler System – Component Description

Oil Cooler

[Refer Figure 79-4](#)

The oil cooler unit is a brazed aluminum plate and corrugated fin assembly. There is a small dimension of 1 mm between the fins to keep to a minimum possible blockage caused by contamination. The external parts of the oil cooler are aluminum alloy. A drain plug is installed at the lowest point on the

oil cooler.

The oil cooler has the environmental limits that follow:

- Altitude (–1000 to 30000 ft.) (–304.8 to 9144 m)
- Ambient Temperature (–65°F to +130 °F) (–54 to 55 °C)
- Oil Temperature (–65°F to +300°F) (–54 to 149 °C)
- System Pressure (375 psig max.) (2586 psig).

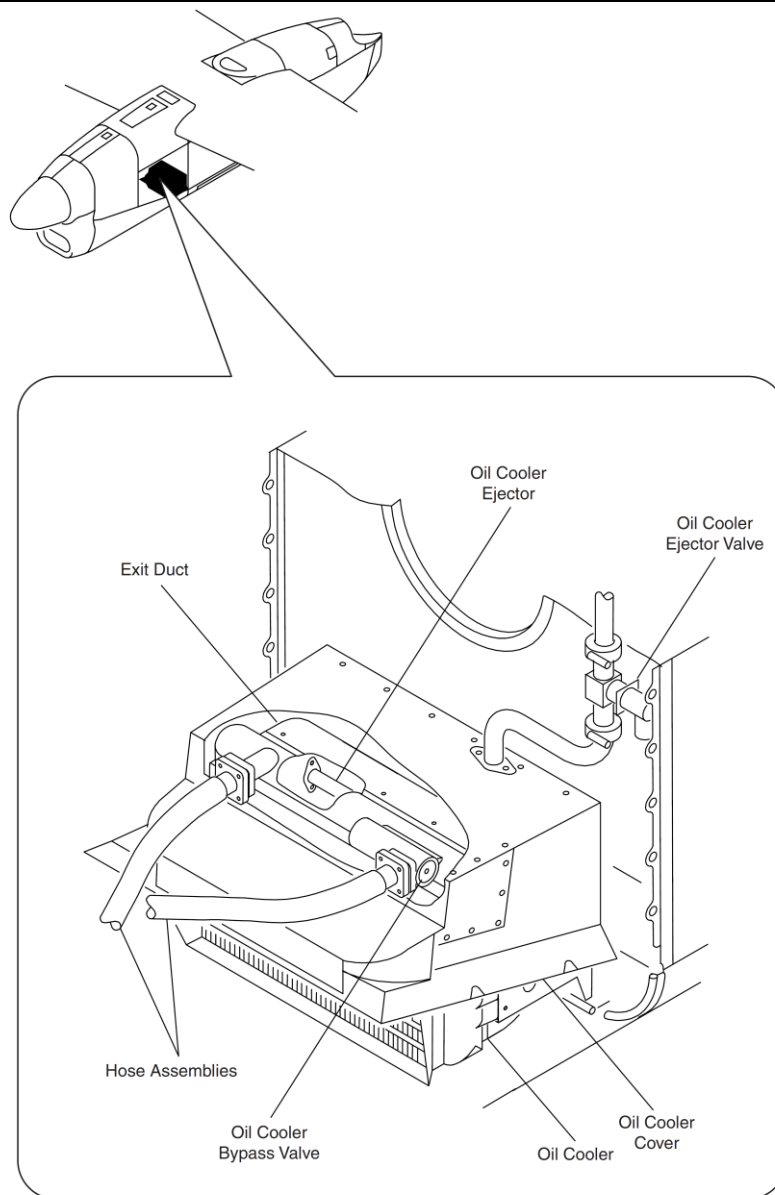


Figure 79-4 Oil Cooler Components

Oil Cooler Bypass Valve

[Refer Figure 79-4](#)

The oil cooler bypass valve is installed in the left side and does the functions that follow:

- A thermal function lets the engine oil fully bypass the oil cooler at temperatures below 170°F (77 °C) (above 185°F (85.0 °C) the engine oil continues to go through the oil cooler)
- A pressure relief function lets the engine oil release if the oil cooler becomes blocked during cold oil operation. The pressure relief operates at 23 psid (158.6 kPad).

Exit Duct

The exit duct is attached with a ten-bolt/anchor nut configuration at the rear of the oil cooler. It is also attached with an eight-bolt/anchor nut configuration at the engine firewall structure. The exit duct is a sheet metal component in a shape satisfactory for airflow. It is also a housing for the oil cooler ejector.

Oil Cooler Ejector

[Refer Figure 79-5](#)

The body of the oil cooler ejector is made of stainless steel and has 22 holes along its length. It is installed in the exit duct, aft of the oil cooler. The oil cooler ejector operates when the aircraft is on the ground. It uses bleed air from downstream of the precooler to eject at high pressure through the 22 holes. This causes air in the nacelle to be pulled through the oil cooler heat-exchanger.

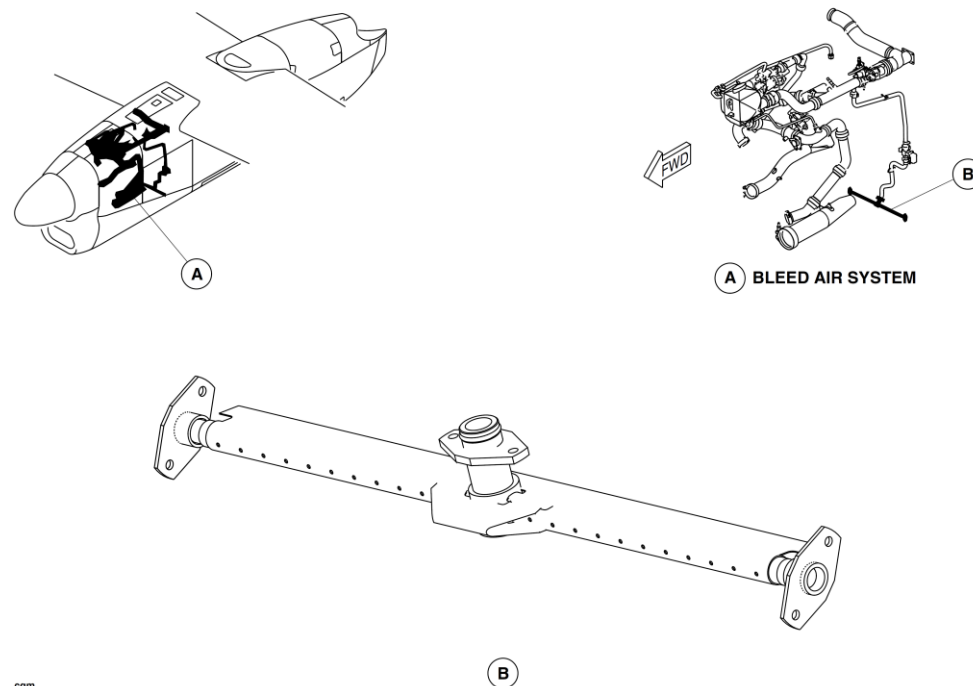


Figure 79-5 Oil Cooler Ejector

Oil Cooler Ejector Valve

[Refer Figure 79-6](#)

The oil cooler ejector valve is installed on the forward face of the engine firewall. A spring holds it in the closed position when the aircraft is on the ground.

An energized solenoid causes the spring to compress and opens the oil cooler ejector valve when the aircraft is on ground.

When the aircraft is airborne a de-energized solenoid causes the spring to retract and closes the oil cooler ejector valve.

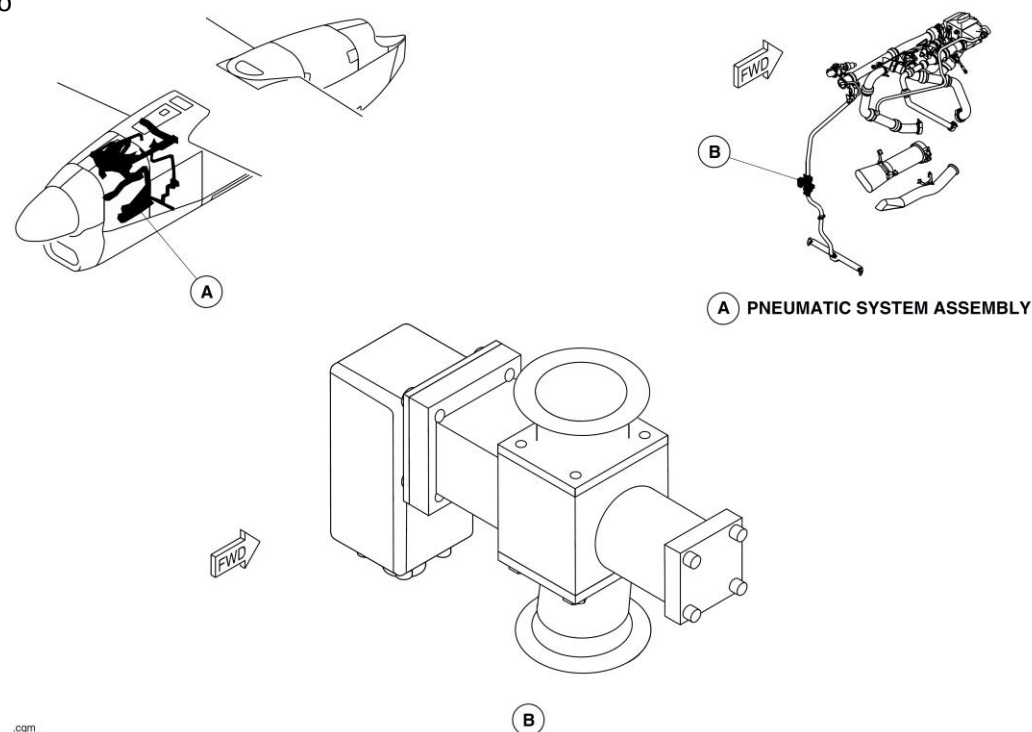


Figure 79-6 Oil Cooler Ejector Valve

Oil Cooler Air Outlet Flap

[Refer Figure 79-7](#)

The oil cooler air outlet flap is in the bottom surface of the nacelle, aft of the engine firewall. It is hinged at its forward end and is operated by an actuator. The oil cooler air outlet flap is fully closed when the engine oil bypasses the oil cooler. When it is necessary for the engine oil to be cooled the oil cooler air outlet flap is open at one of three positions. This lets the necessary airflow release to the atmosphere.

When the oil temperature is more than the limit on the ground, the oil cooler air outlet flap is fully open at 32 degrees. The oil cooler air outlet flap is fully closed when the oil temperature is less than the limit on the ground. When the oil temperature is more than the limit when airborne, the oil cooler air outlet flap is open at 5 degrees. When the engine operation conditions are unusual when airborne and the oil temperature is more than the limit, the oil cooler air outlet flap is open at 10 degrees.

Oil Cooler Air Outlet Flap Actuator

[Refer Figure 79-7 & 79-8](#)

The oil cooler air outlet flap actuator is attached to the nacelle structure and the oil cooler air outlet flap. It receives signals from the Full Authority Digital Engine Computer (FADEC), through the Engine Control Interface Unit (ECIU). The signals give instruction to fully open or close the oil cooler air outlet flap (with two open positions between, if necessary). The time for the travel of the oil cooler air outlet flap actuator with a voltage supply of 26 Vdc is as follows:

- From 0 to 32 degrees on the ground (5 to 8 sec.)
- From 0 to 5 degrees when airborne (1.5 to 3 sec.)
- From 5 to 10 degrees when airborne (1.5 to 3 sec.)
- From 0 to 10 degrees when airborne (3 to 6 sec.)
- From 32 to 0 degrees when airborne (less than 8 sec.).

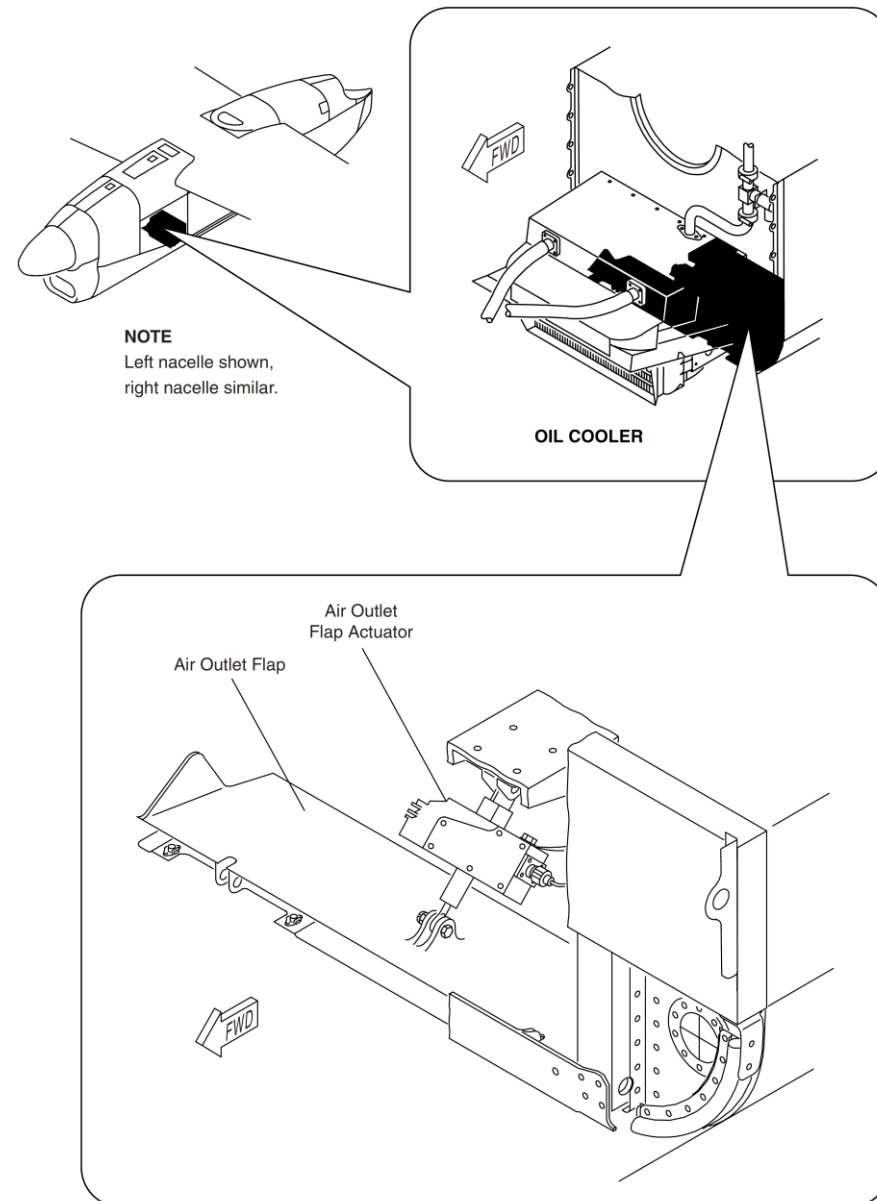


Figure 79-7 Oil Cooler Air Outlet Flap & Actuator

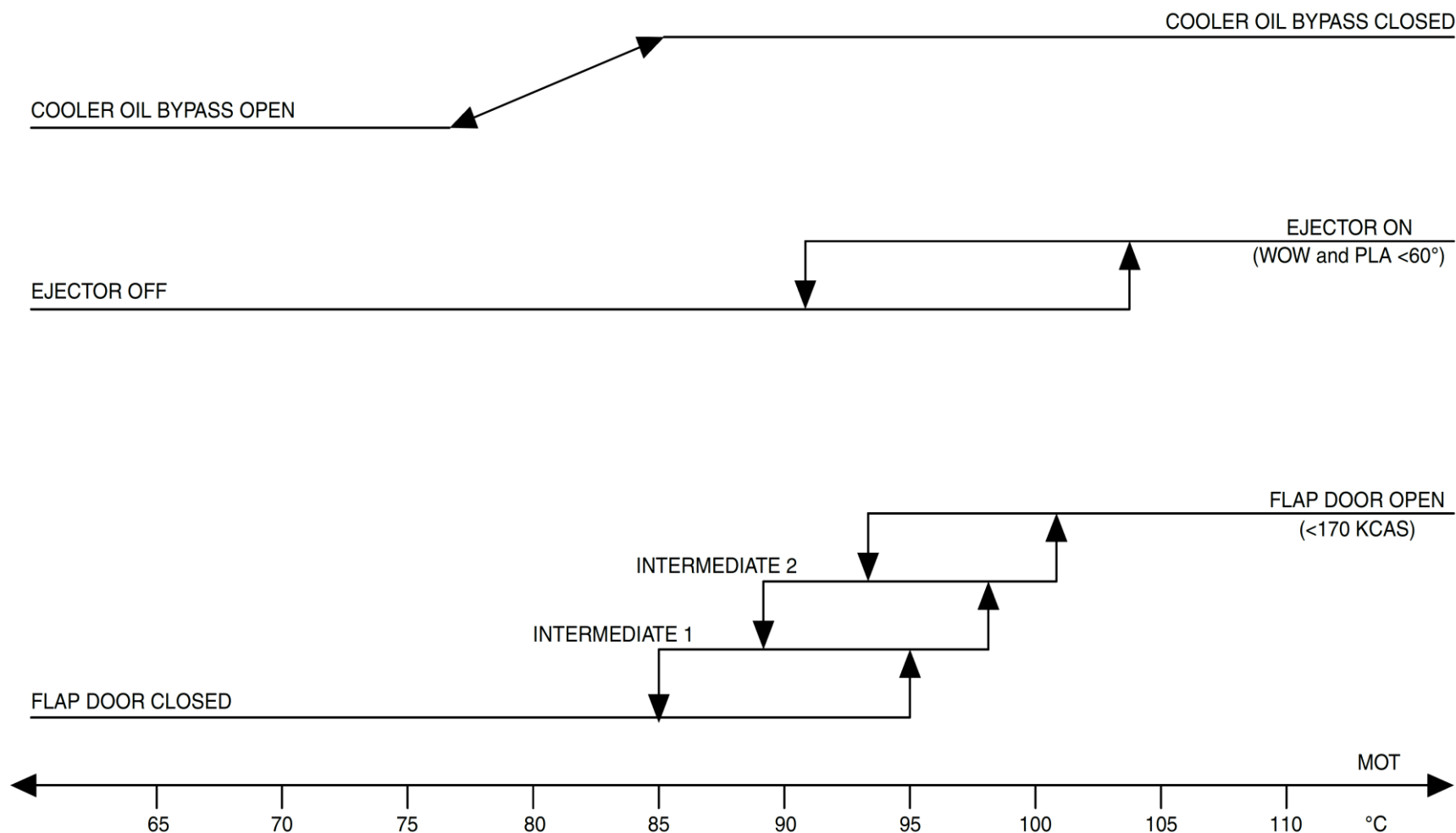


Figure 79-8 Oil Cooler Air Outlet Flap Operation Detail

Oil Pressure Indicating System

Introduction

The purpose of the Oil Pressure Indicating System is to provide the oil pressure status to the flight compartment.

General Description

[Refer Figure 79-9](#)

The Oil Pressure Indicating System uses a transducer to determine and

transmit the engine oil pressure status to the flight compartment.

Detailed Description

The Oil Pressure Indicating System uses a transducer to convert the oil pressure into an electrical signal and compare it to the No. 5 bearing cavity air pressure. An electrical circuit in the transducer outputs a signal to the flight compartment.

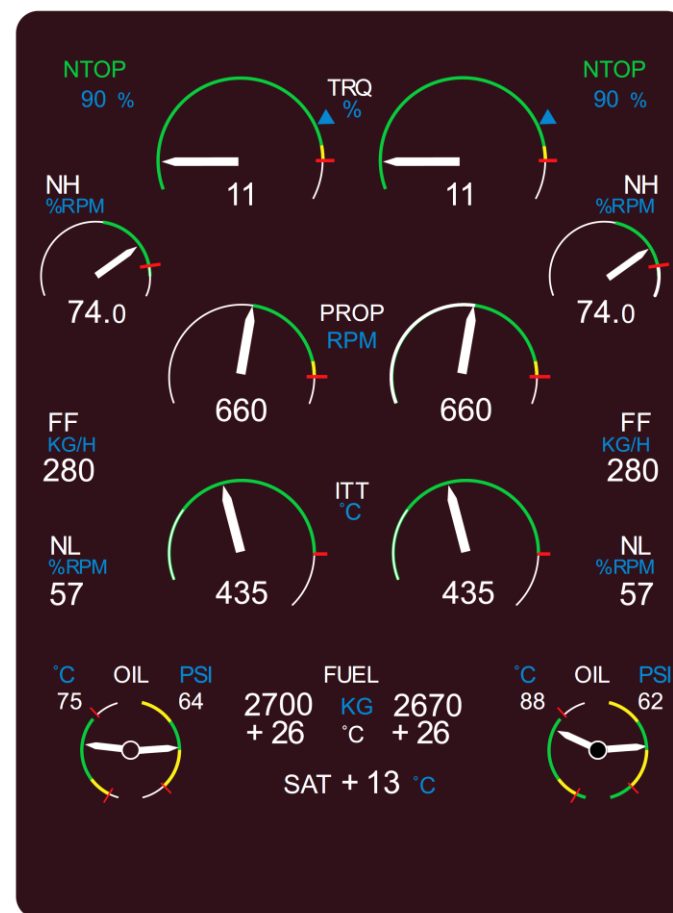
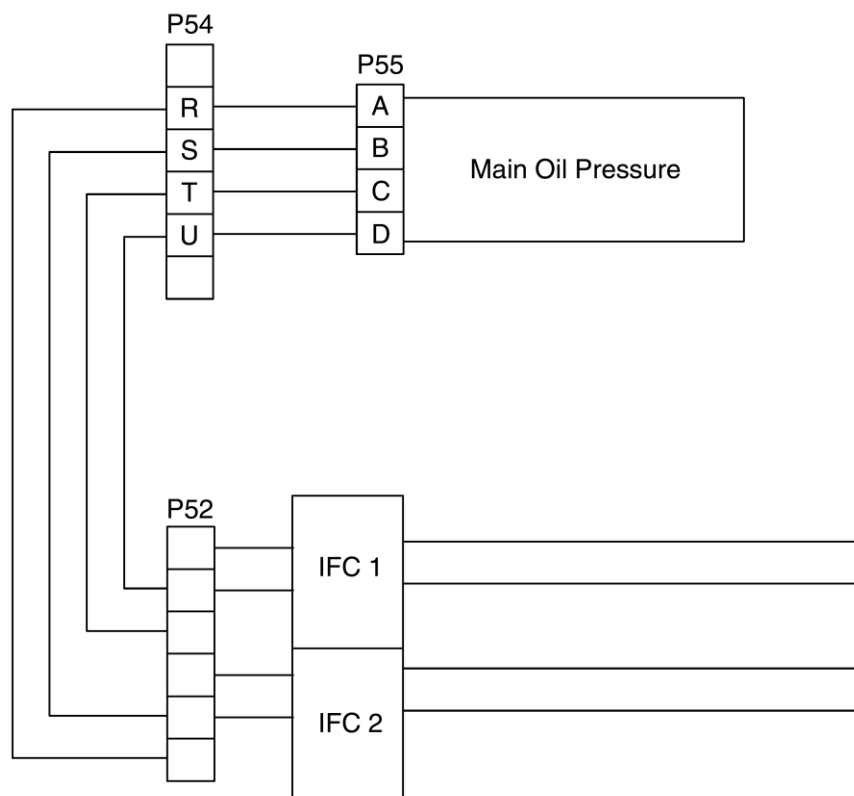


Figure 79-9 Oil Pressure Indication Schematic

Oil Pressure Indicating System – Component Description

Oil Pressure Transducer

[Refer Figure 79-10](#)

The oil pressure transducer is located at the bottom of the low pressure compressor case on the left side of the engine. It is installed in an oil wetted cavity and is held in place with two bolts and sealed with preformed packings. The P55 wiring harness connector is connected to the transducer. This connector transmits the transducer signal to the flight deck through the wiring harness. A ground strap connects the housing of the transducer to the low pressure compressor case.

The oil pressure transducer has a housing, an inlet port and a reference port, two pressure sensing elements, an electrical circuit board and a connector. The elements have two channels to provide for a backup if one channel fails. The interior of the transducer is filled with foam to prevent moisture ingress. The transducer is a welded assembly and cannot be disassembled in the field.

During engine operation, oil enters the inlet port and pushes on the pressure sensing element. The element converts the oil pressure to an electrical signal. Air from the No. 5 bearing cavity enters the reference port and pushes on the second pressure sensing element. This element converts the air pressure to an electrical signal. The two electrical signals are processed by the electrical circuitry and a signal is output and transmitted through the electrical wiring harness to the flight deck.

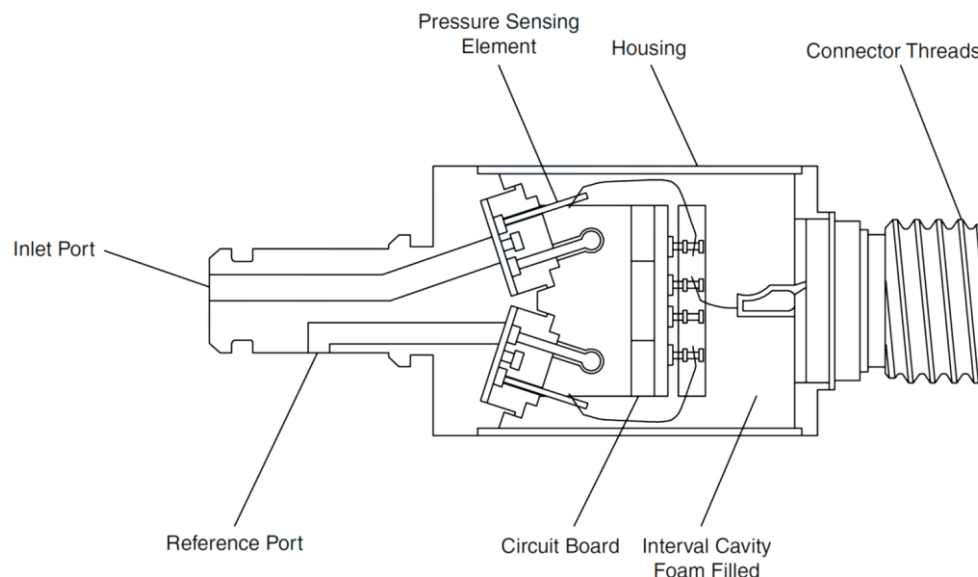


Figure 79-10 Oil Pressure Transducer

Oil Temperature Indicating System

Introduction

The Oil Temperature Indicating System supplies oil temperature status to the Full Authority Digital Electronic Control (FADEC).

General Description

The Oil Temperature Indicating System uses a bi-metallic junction sensor to convert the oil temperature and Indicated Turbine Temperature (ITT) into electrical signals and transmit these signals to the FADEC.

Detailed Description

[Refer Figure 79-11](#)

The Oil Temperature Indicating System uses a Main Oil Temperature (MOT) sensor to determine and transmit engine oil temperature signals to the FADEC. The FADEC processes these signals and outputs an indication to the flight compartment. The FADEC also uses these signals, and the signal from the thermocouples, to calculate the ITT temperature.

In normal conditions, the green band on the oil temperature indication at the engine instrument panel, shows a temperature of 0–107 °C (32–225 °F). In icing conditions, when the prop de-ice switch is on, the green band contracts and it shows a temperature range of 65–107 °C (149–225 °F).

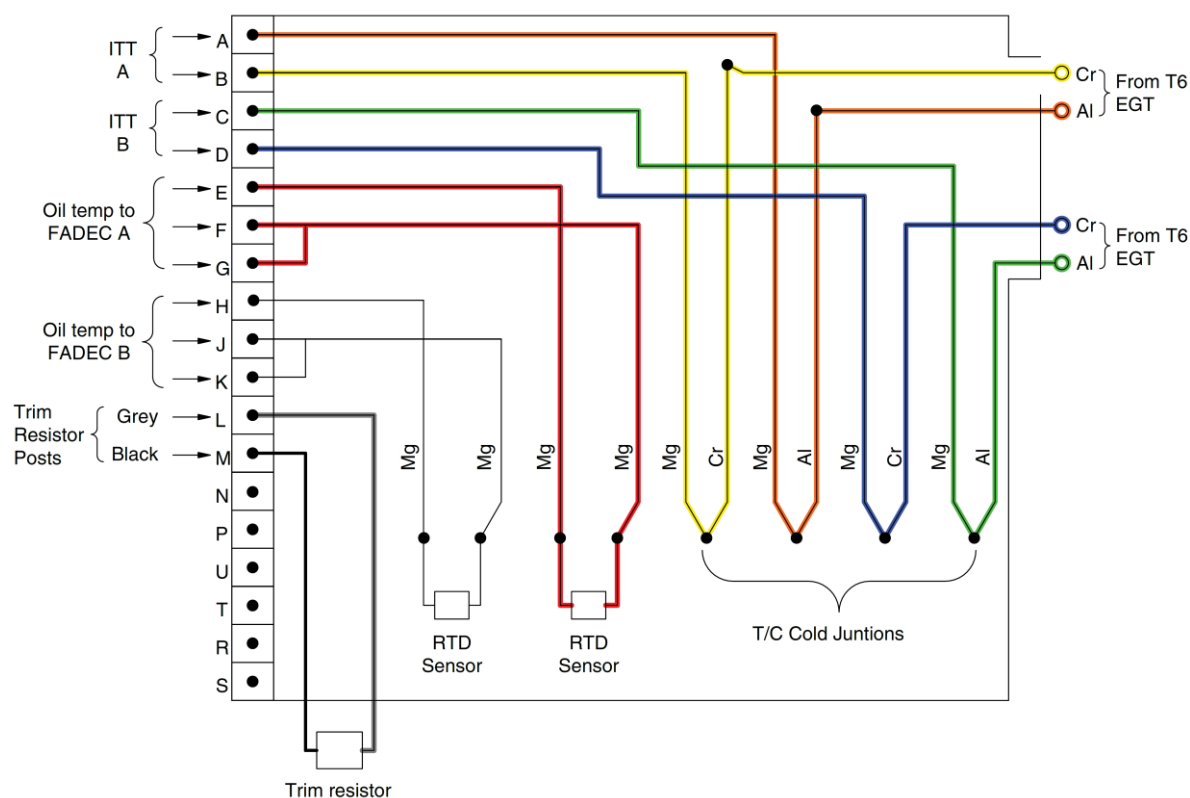


Figure 79-11 Oil Temperature Indicating System Schematic

Oil Temperature Indicating System – Component Description

Oil Temperature Sensor

Refer Figure 79-12

The oil temperature sensor is located on the top of the air inlet case. It is installed in an oil wetted cavity and is held in place with two bolts and sealed with a preformed packing. The P16 wiring harness connector is connected to a receptacle mounted on the sensor. This connector transmits the sensor signals to the FADEC through the wiring harness. A ground strap connects the housing of the sensor to the air inlet case.

The sensor has a housing, and four conductors of different material, two Chromel and two Alumel, that are joined at one end (bimetal junction). This gives the sensor two channels to provide for a backup if one channel fails.

The other ends of the four Chromel and Alumel conductors, are connected to the receptacle and to a flexible cable that goes out of the sensor. The other end of the cable connects to the ITT thermocouple wiring harness with four terminals that have an anti-rotation property.

On the outside of the sensor are two mounting studs. These studs hold the ITT Trim Resistor. The FADEC uses the resistor to trim the ITT signal to compensate for engine to engine differences in measured ITT and to keep all engines within a small ITT window.

The sensor is a welded assembly and cannot be disassembled in the field.

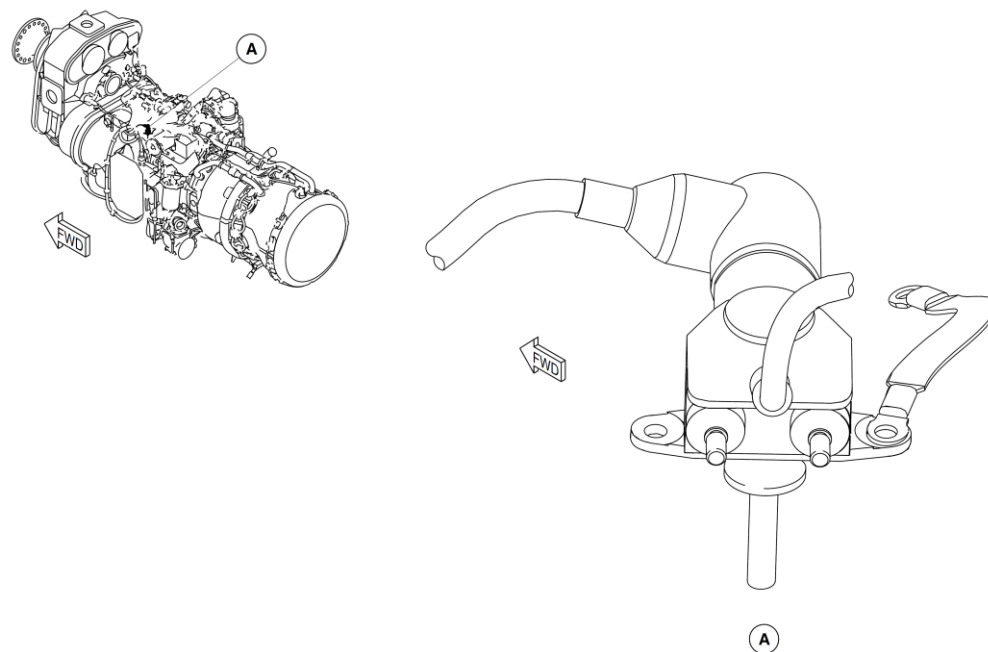


Figure 79-12 Oil Temperature Sensor

Low Oil Pressure Warning System

Introduction

The Low Oil Pressure Warning System supplies a low oil pressure warning signal to the Full Authority Digital Electronic Control (FADEC).

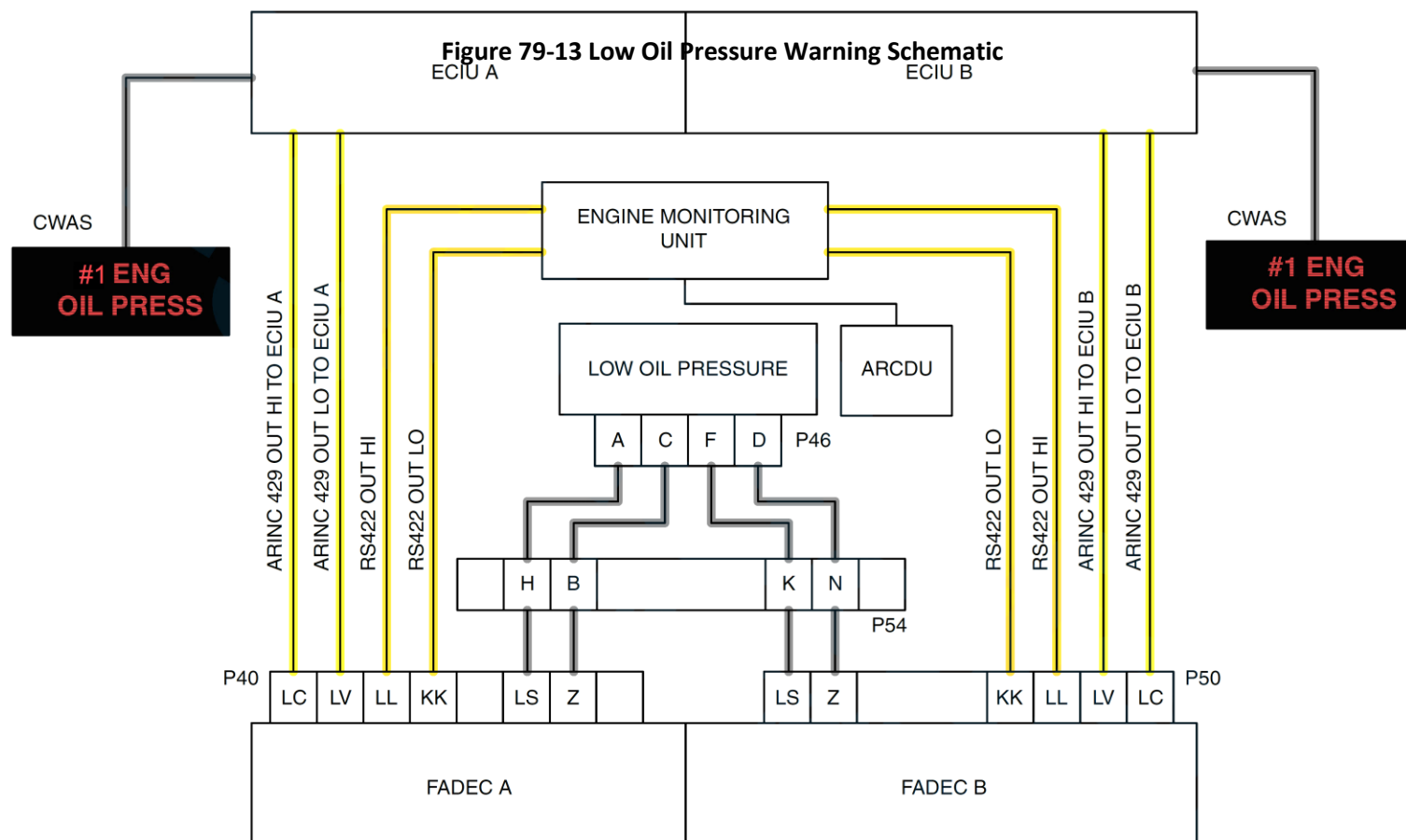
General Description

The Low Oil Pressure Warning System uses a switch to determine a low oil pressure condition.

Detailed Description

[Refer Figure 79-13](#)

The Low Oil Pressure Warning System uses a switch to convert the low oil pressure into an electrical signal. This signal is sent to the FADEC which then sends an electrical signal to the flight compartment and the Engine Monitoring System (EMS).



Low Oil Pressure Warning System – Component Description

Low Oil Pressure Switch

[Refer Figure 79-14](#)

The Low Oil Pressure Switch is located at the bottom of the low pressure compressor case on the port side of the engine. It is installed in an oil wetted cavity and is held in place with two bolts and sealed with preformed packings. The P46 wiring harness connector is connected to the switch. This connector transmits the switch signal to the FADEC through the wiring harness. A ground strap connects the housing of the switch to the low pressure compressor case.

The Low Oil Pressure Switch has a housing, an inlet port and a reference port, a mechanical switch, and a connector. The switch has two channels to provide for a backup if one channel fails. The switch is a welded assembly and cannot be disassembled in the field.

During engine operation, oil enters the inlet port and holds the dual channel pressure switch open. If the oil pressure decreases to 44 ± 3 psi (303 ± 21 kPa) the switch will close and a signal is sent to the FADEC. The FADEC then sends a signal to the flight compartment and the EMS.

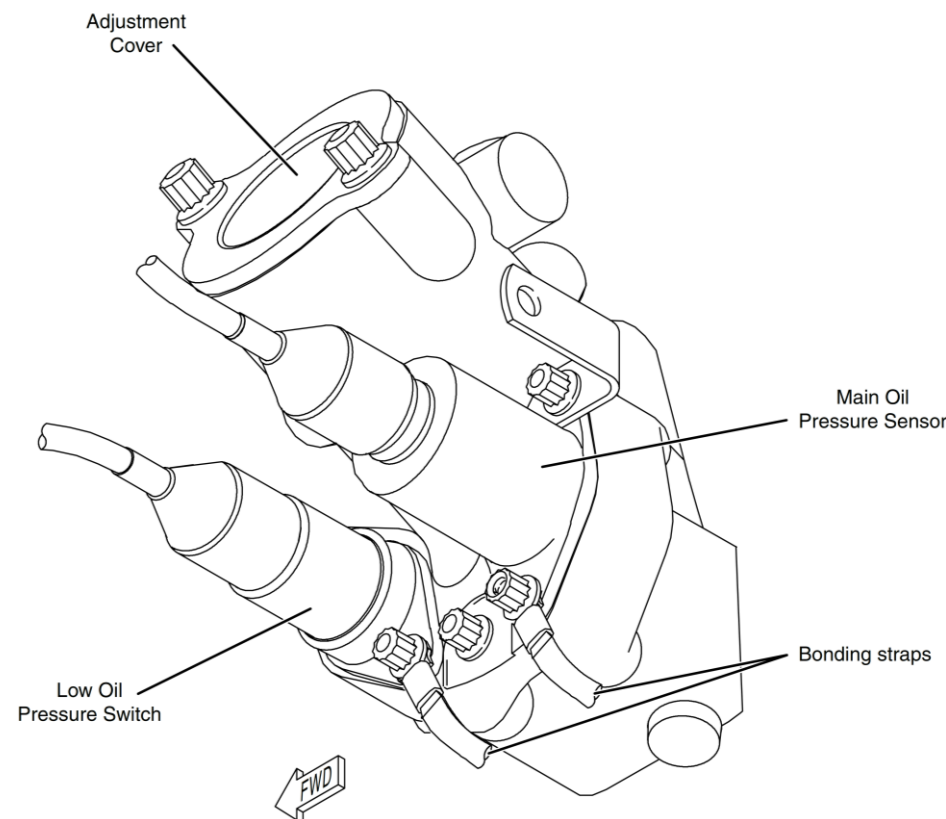


Figure 79-14 Low Oil Pressure Switch

Low Oil Pressure Warning Indication – Electrical Schematic

[Refer Figure 79-15](#)

During engine operation if the oil pressure drops to <44 psid, a signal will be sent from the low oil pressure switch to FADEC EEC:

- The signal will enter FADEC EEC on Pin S of both channels
- Channel NOT in command would output data on ARINC 429 Pins C and V to the ECIU A at Pins 13 and 14, and Pins 91 and 92 of ECIU B.
- ECIU B will remove the ground on Pin 97 to bring ON the #2 ENG OIL PRESS warning light.
- FADEC EEC will output data on RS 422 bus through Pins LL and KK to the EMU on Pins D3 and C3.
- EMU will output data on Pins A11 and B11 for fault code display on ARCDU through the IOP.

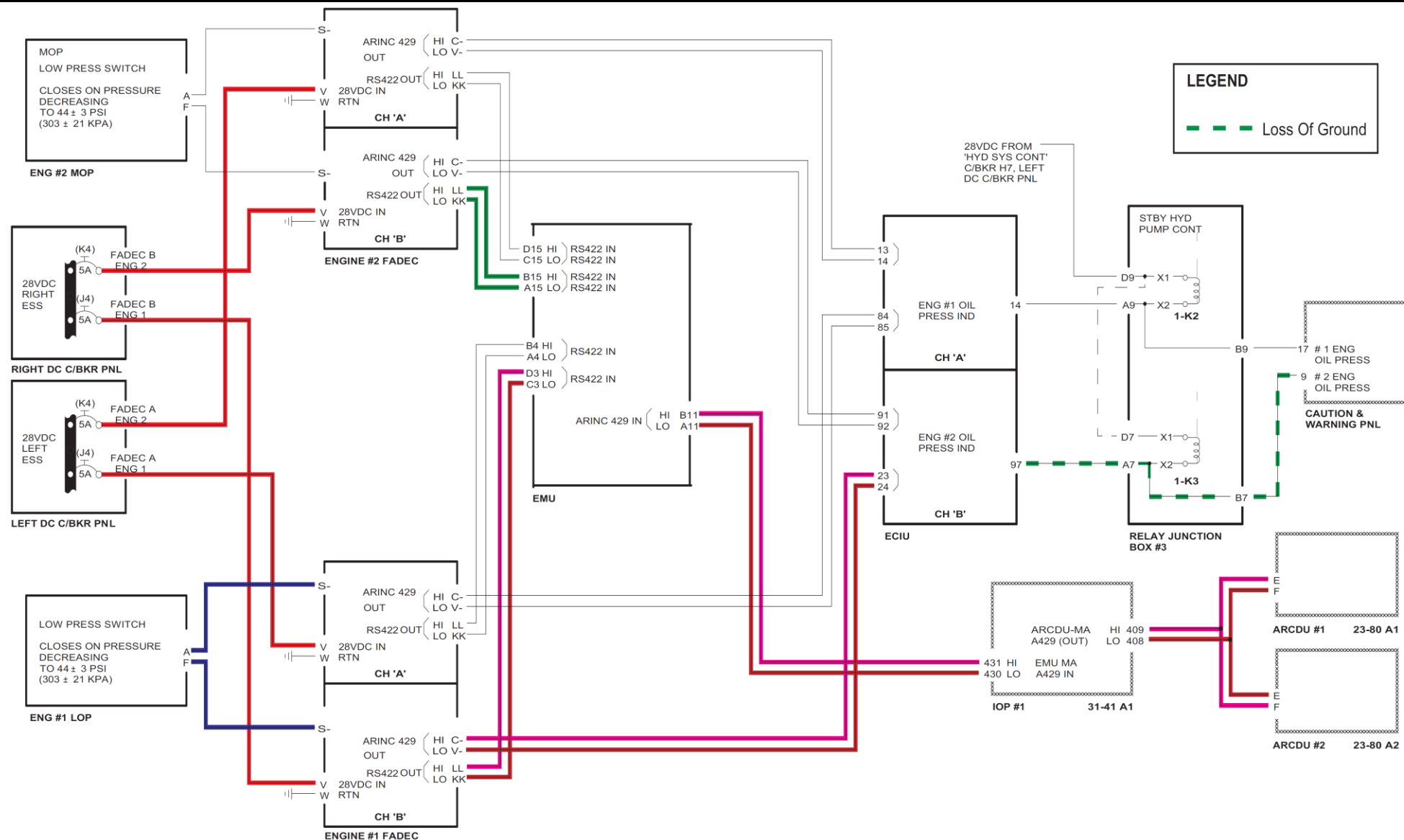


Figure 79-15 Low Oil Pressure Warning – Electrical Schematic

Chip Detection System

General

[Refer Figure 79-16](#)

The chip detection system uses three chip detectors to indicate the presence of ferrous chips coming from the:

- AC generator
- Turbomachinery module
- RGB module.

System Description

The system signals are transmitted electrically to the FADEC channel A and transferred internally to FADEC channel B.

Ferrous debris in the oil system is attracted to the chip detector poles.

If the poles are bridged (by debris), a signal is sent to the FADEC. The FADEC will then output a RS422 signal to the EMU.

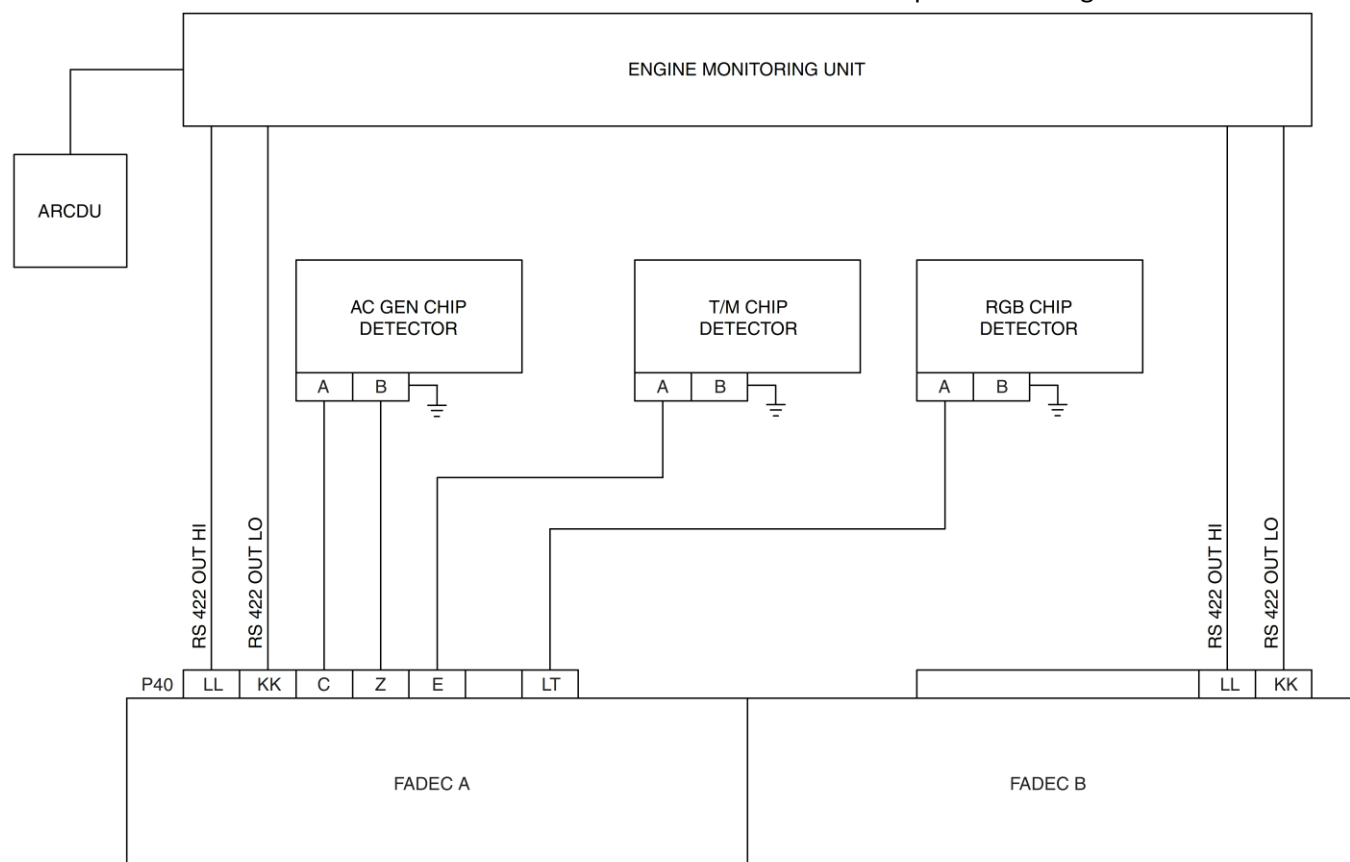


Figure 79-16 Chip Detection System - Schematic

Oil Filter Bypass Warning System

General

[Refer Figure 79-17](#)

The oil filter impending bypass warning system warns of an impending bypass of oil across the oil filter. This indicates a contaminated oil filter.

System Description

When the oil filter differential pressure exceeds 22 psid (152 kPad), the pressure switch opens.

This state change is detected by the FADEC, which in turn sends a discrete signal to the EMU.

The EMU makes a record of a main oil filter impending bypass event.

During operation with cold oil (main oil temperature less than 175 °F (80°C)), any impending bypass signal is ignored by the FADEC.

This will prevent false event messages from being stored in the EMU.

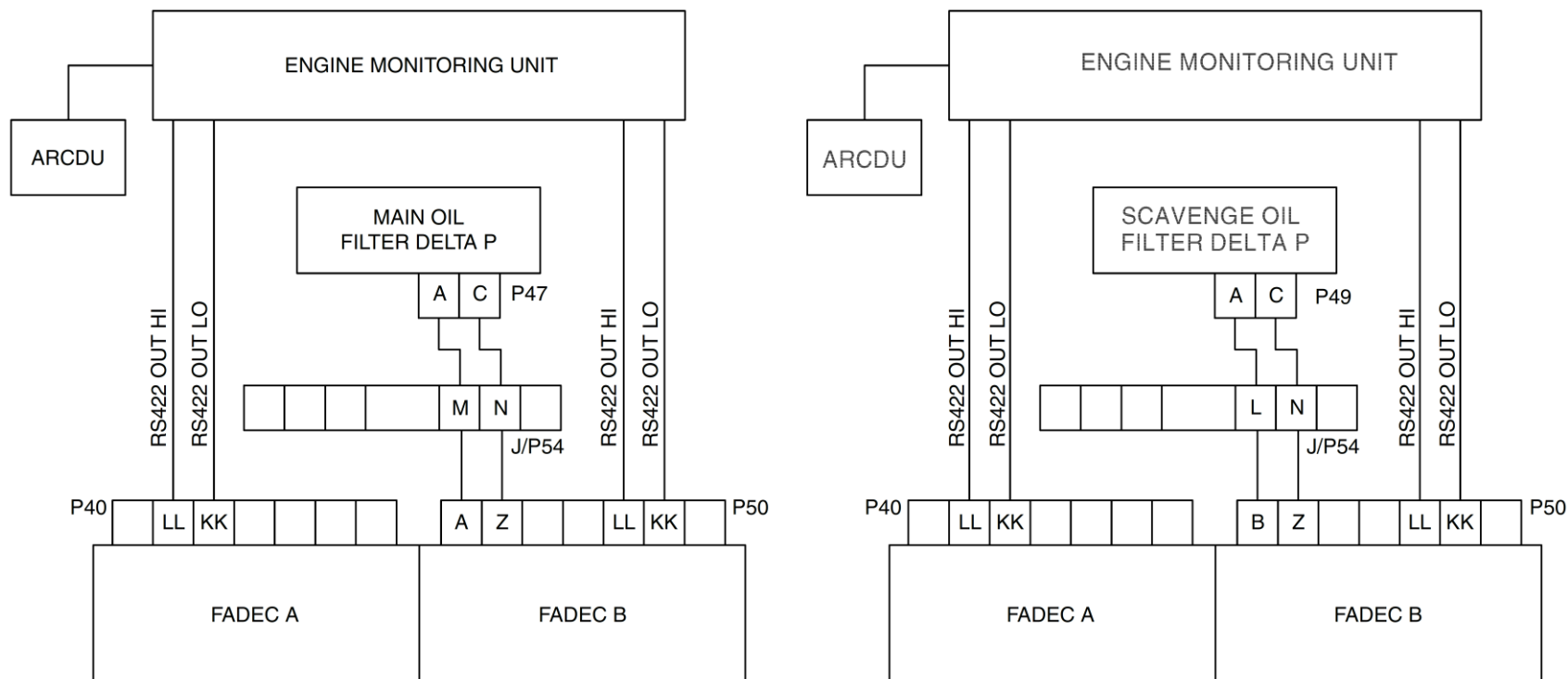


Figure 79-17 Oil Filter Impending Bypass Warning System - Schematic

Oil Filter Bypass Warning System – Component Description

Oil Filter Impending Bypass Switch

[Refer Figure 79-18](#)

There are two impending bypass switches in the system.

One is located on the main oil filter housing; the other on the scavenge oil filter housing. They are installed in oil wetted cavities that are attached with bolts and sealed with preformed packings.

The oil filter impending bypass switch detects the pressure drop across a partially blocked engine oil filter and sends a signal to FADEC.

Wiring harness connectors (P47 and P49) connect to the switches. The connectors transmit the switch signals to the FADEC channel B which sends it internally to channel A. Ground straps connect the switch housings of the transducer to the engine cases.

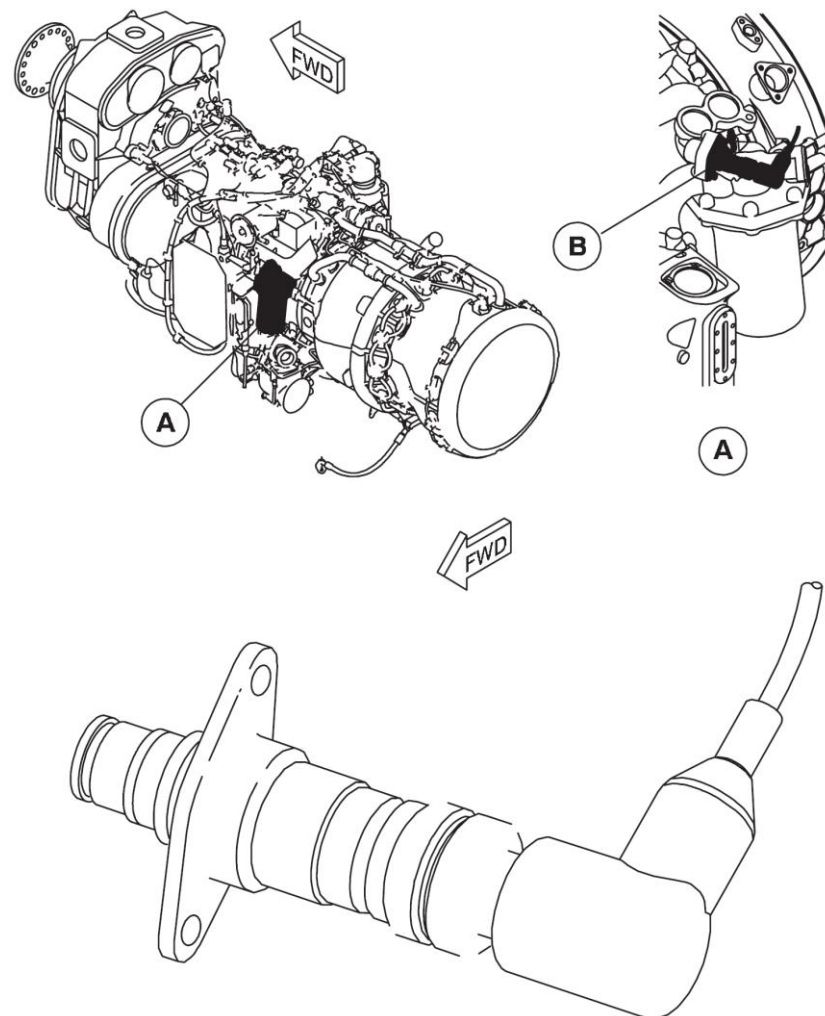


Figure 79-18 Oil Filter Impending Bypass Switch

Oil Quantity Indicating System

Introduction

[Refer Figure 79-19](#)

On the engine, the quantity of oil that is in the tank is measured with three components:.

- a calibrated dipstick
- an oil level indicator sight glass
- A low oil level indicator.

These three quantity-indicating components are located near each other, on the oil tank, on the left side of the engine.

General Description

During normal engine operation, oil is consumed and must be replaced during the next servicing period. Since not all the oil is consumed, the maintenance crew needs to know how much oil to add to the oil tank. The oil quantity can be determined with a calibrated dipstick, an oil level indicator sight glass and a low oil level indicator glass.

Oil Quantity Indicating System – Component Description

Calibrated Dipstick

The calibrated dipstick is attached to the oil filler cap. The calibrated markings give indications on the quantity of oil that is to be added to the oil tank. These marked calibrations are shown as follows:

- ADD 1 LTR
- ADD 2 LTR
- ADD 3 LTR
- MIN HOT.

Oil Level Indicator Sight glass

The oil level indicator sight glass is attached to the oil tank with two bolts and sealed with preformed packings. The indicator sight glass has calibrated markings to give indications of the quantity of oil that is to be added to the oil tank. These marked calibrations are shown as follows:

- MAX HOT
- OIL
- ADD 1 US QT
- ADD 1 LTR

Low Oil Level Indicator Glass

The low oil level indicator glass is located below the oil level indicator sight glass. The low oil level indicator glass is threaded into the oil tank and is sealed with a preformed packing. This indicator gives a visual indication that the quantity level of oil in the tank is in a low condition.

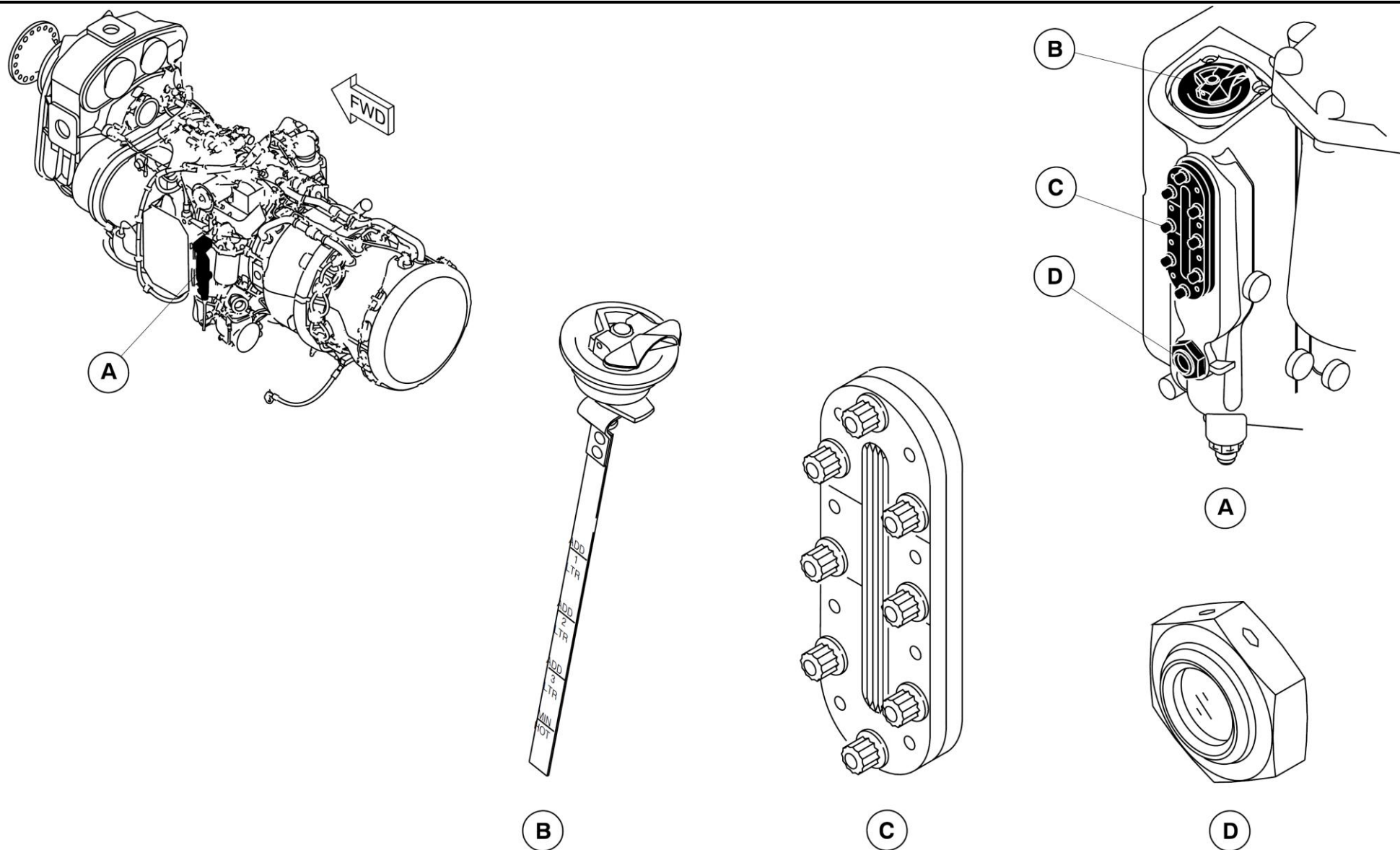


Figure 79-19 Oil Quantity Indication

Remote Oil Level Indication System (Optional)

Introduction

The remote oil level indication system is installed on the Dash 8–400 as an option under CR812CH00006 (written as “system” in this description procedure). The system is non–essential for the operation of the aircraft.

The system gives indication that there is approximately 10 flying hours or less of engine oil (based on normal oil consumption).

The Aircraft Maintenance Manual (AMM) still requires the oil level check using the sight glass and low oil level indicator.

System Description

[Refer Figure 79-20](#)

The system cockpit display consists of two green lights (one for each engine as indicated) and a system interrogation switch with three positions, OFF, LAMP TEST AND LEVEL CHECK. The control panel is located on the pilot’s forward side console.

When the system interrogation switch is selected to “LAMP TEST,” the ENG 1 and ENG 2 has a steady light.

When the system interrogation switch is selected to “LEVEL CHECK,” the ENG 1 and ENG 2 oil level indications are as follows:

- Light steady, oil level is good.
- Light off, low oil level or 10 hours (or less) flight time remaining.
- Light flashing, no valid reading or EMU is being interrogated by the Central Diagnostic System (CDS).

The standard engine oil filler cap/dipstick is replaced by an oil filler cap/sensor assembly. The sensor is a reed type sensor switch set at 10 hours above the oil level where the loss of oil pressure may occur if operating at the top of the engine limits.

FADEC software monitors the oil level reed switch and sends oil level

condition to the Engine Monitoring Unit (EMU).

The EMU performs the required logic, verification and monitoring of the engine operation to determine the oil level status.

The EMU checks for oil level status:

- 10 seconds after electrical power up of the FADEC and EMU.
- 60 seconds after engine shutdown.

At electrical power up, EMU oil level status is updated if both:

- Oil temperatures is less than 88°C.
- Reed switch position has been stable for 10 seconds.

At engine shut down, oil level status is updated:

- When engine is running for longer than 15 minutes.
- When the engine is operated between Disc and Flight Idle for more than 2 minutes before shut down (allow for full oil scavenging).
- When the engine has been shut down for 1 to 15 minutes (electrical power to FADEC is “ON”).
- Main oil temperature (MOT) is less than 88°C.
- Reed switch position is stable for a minimum 10 seconds.

The EMU will allow for an update of the switch status if it has changed from open to close indicating that the oil tank has been replenished with the engine running, stabilized idle and shutdown time requirements are ignored.

A flashing indicator lamp means the EMU has been unable to achieve a successful update of the switch position after more than 2.5 hours of engine running time since the last update or the EMU is being interrogated through the CDS.

The EMU Oil Level Status is stored in the EMU so that the Oil Level Indicating System can be retrieved at any time (even in–flight) and the control panel is placarded “FOR GROUND TEST ONLY/ENG OFF.”

Any of the following indications is given when the system interrogation switch is selected to “LEVEL CHECK.”

Condition	Flight Compartment Display
Oil Level OK	Steady Green Light
Oil Level Low	No Light
No Valid Reading	Flashing Green Light
EMU Switch Failed	No Light
FADEC Channel Failed	No Light

Table 79-1 Remote Oil Indications Chart

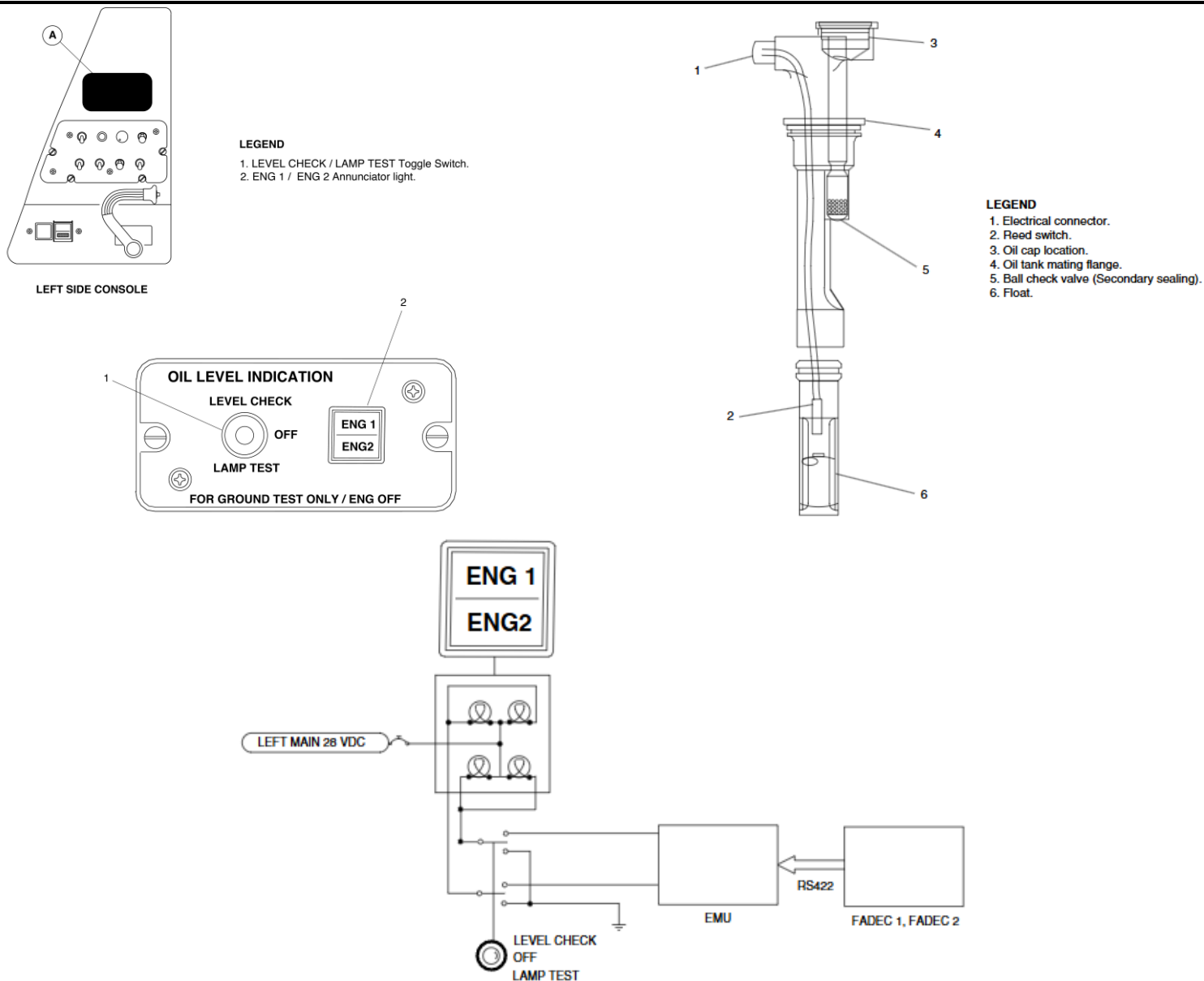


Figure 79-20 Remote Oil Level Indication System